

Tong Statistical Field Theory — Sheet 1: Landau Theory

D. Tong, Statistical Field Theory, Example Sheet 1; 8 worked problems.

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These prompts paraphrase David Tong's Statistical Field Theory, Example Sheet 1 (October 2018). Problem numbers, starred status, equations, and guidance follow the official sheet.

1. Sheet 1, Problem 1: Transfer Matrix for the One-Dimensional Ising Model

Take N Ising spins with $s_{N+1} = s_1$ and partition function

$$Z = \sum_{s_1=\pm 1} \cdots \sum_{s_N=\pm 1} \prod_{i=1}^N \exp \left[\beta J s_i s_{i+1} + \frac{\beta B}{2} (s_i + s_{i+1}) \right]. \quad (1.1)$$

Show that $Z = \text{tr } T^N$ for the transfer matrix

$$T = \begin{pmatrix} e^{\beta J - \beta B} & e^{-\beta J} \\ e^{-\beta J} & e^{\beta J + \beta B} \end{pmatrix}. \quad (1.2)$$

Hint: Express the i th factor as $T_{s_i, s_{i+1}} = \exp[\beta J s_i s_{i+1} + \beta B(s_i + s_{i+1})/2]$, with s_i labelling a column and s_{i+1} a row. Find the two eigenvalues,

$$\lambda_{\pm} = e^{\beta J} \cosh(\beta B) \pm \sqrt{e^{2\beta J} \cosh^2(\beta B) - 2 \sinh(2\beta J)}, \quad (1.3)$$

and explain why $Z \simeq \lambda_+^N$ as $N \rightarrow \infty$. At $B = 0$, deduce that the magnetisation is always zero and that temperature causes no phase transition in this one-dimensional system.

Solution. The partition function can be written as

$$Z = \sum_{s_1=\pm 1} \cdots \sum_{s_N=\pm 1} T_{s_1, s_2} T_{s_2, s_3} \cdots T_{s_N, s_1} = \text{tr } T^N = \lambda_+^N + \lambda_-^N. \quad (1.4)$$

Clearly, $\lambda_+ > \lambda_-$. Therefore, for Large N , we can write $Z \simeq \lambda_+^N$. The magnetisation is given by

$$M = \frac{1}{\beta} \frac{\partial \log Z}{\partial B} = \frac{1}{\beta Z} \left[N \lambda_+^{N-1} \frac{\partial \lambda_+}{\partial B} + N \lambda_-^{N-1} \frac{\partial \lambda_-}{\partial B} \right]. \quad (1.5)$$

Clearly, $\left. \frac{\partial \lambda_{\pm}}{\partial B} \right|_{B=0} = 0$, which gives $M \Big|_{B=0} = 0$. Therefore, there is no second-order phase transition for a one-dimensional system.

2. Sheet 1, Problem 2: Weiss Mean Field Theory

Expand an Ising nearest-neighbour product about the equilibrium magnetisation m as

$$s_i s_j = (s_i - m)(s_j - m) + m(s_i - m) + m(s_j - m) + m^2. \quad (2.1)$$

Neglect the product of fluctuations. For the source's nearest-neighbour parameter q , derive

$$Z = e^{-\beta J N q m^2 / 2} 2^N \cosh^N(\beta J q m + \beta B), \quad (2.2)$$

and obtain the self-consistency equation

$$m = \tanh(\beta B + \beta J q m). \quad (2.3)$$

At $B = 0$, compare the graphical solutions above and below $T_c = Jq$. For nonzero B , find the high-temperature asymptotic solution.

Solution. Here, we shall treat m merely as the equilibrium distribution. Keeping terms linear in fluctuation, we have

$$s_i s_j \approx m(s_i - m) + m(s_j - m) + m^2 = m(s_i + s_j) - m^2. \quad (2.4)$$

Then,

$$\sum_{\langle ij \rangle} s_i s_j \approx -\frac{N q m^2}{2} + q m \sum_i s_i, \quad (2.5)$$

which gives

$$E[s_i] \approx -(B + J m q) \sum_i s_i + \frac{N J q m^2}{2} = -(B + J m q) \bar{s} + \frac{N J q m^2}{2}, \quad (2.6)$$

where $\bar{s} = \sum_i s_i$. Summing over all spin configurations (including the binomial multiplicity of each value of $\bar{s}$), the partition function factorises:

$$Z \approx e^{-\beta J N q m^2 / 2} \prod_{i=1}^N \sum_{s_i = \pm 1} e^{\beta(B + J q m) s_i} = e^{-\beta J N q m^2 / 2} 2^N \cosh^N(\beta J q m + \beta B). \quad (2.7)$$

The factorisation is exact within the mean-field approximation for either parity of N . The self-consistency equation immediately follows from

$$m = \frac{1}{N\beta} \frac{\partial \log Z}{\partial B}. \quad (2.8)$$

Solving this equation would give us the allowed value for m . If we differentiate the RHS, we find

$$\frac{\beta J q}{\cosh^2(\beta J q m)} \leq \beta J q. \quad (2.9)$$

For $T > T_c$, $\beta J q < 1$, and hence

$$\frac{\beta J q}{\cosh^2(\beta J q m)} \leq 1, \quad (2.10)$$

which means that $m = 0$ is the only solution to the self-consistency equation. However, for $T < T_c$, there can be two extra solutions to the equation. This is a continuous (second-order) phase transition: the stable nonzero solutions approach zero as $T \rightarrow T_c^-$.

For fixed nonzero B and $T \rightarrow \infty$, linearising the self-consistency equation gives $m \simeq B/(T - Jq) \sim B/T$ (with $k_B = 1$).

3. Sheet 1, Problem 3: Exact Mean Field Limit of a Long-Range Ising Model

Let every spin interact with every other spin, including itself, through

$$E = -B \sum_i s_i - \frac{J}{2N} \left(\sum_i s_i \right) \left(\sum_j s_j \right). \quad (3.1)$$

Explain the factor $1/N$. Establish the Gaussian identity

$$e^{\beta J \alpha^2 / (2N)} = \sqrt{\frac{N\beta J}{2\pi}} \int_{-\infty}^{\infty} dx e^{-N\beta J x^2 / 2 + \alpha\beta J x}, \quad (3.2)$$

and use it to express the partition function as

$$Z = \sqrt{\frac{N\beta J}{2\pi}} \int_{-\infty}^{\infty} dx e^{-NS(x)}, \quad S(x) = \frac{\beta J x^2}{2} - \log[2 \cosh(\beta(B + Jx))]. \quad (3.3)$$

Find the saddle x_* that dominates at large N , up to its prefactor. Derive its stationarity equation and show that $m = x_*$.

Solution. Since we expect the energy to scale as N , while $\sum_i s_i$ also scales as N . Therefore, we must have a factor of $1/N$.

The Gaussian identity is straightforward. Let us define

$$\sum_i s_i \equiv \alpha. \quad (3.4)$$

Then,

$$\begin{aligned} Z &= \sum_{\alpha} e^{\beta J \alpha^2 / (2N) + \beta B \alpha} \\ &= \sqrt{\frac{N\beta J}{2\pi}} \sum_{\alpha} e^{\beta B \alpha} \int_{-\infty}^{\infty} dx e^{-N\beta J x^2 / 2 + \alpha\beta J x} \\ &= \sqrt{\frac{N\beta J}{2\pi}} \int_{-\infty}^{\infty} dx e^{-N\beta J x^2 / 2} \sum_{\alpha} e^{\beta(B + Jx)\alpha} \\ &= \sqrt{\frac{N\beta J}{2\pi}} \int_{-\infty}^{\infty} dx e^{-N\beta J x^2 / 2} \cdot 2^N \cosh^N \beta(B + Jx) \\ &= \sqrt{\frac{N\beta J}{2\pi}} \int_{-\infty}^{\infty} dx e^{-NS(x)}. \end{aligned} \quad (3.5)$$

The saddle is given by

$$S'(x_*) = 0 \implies x_* = \tanh(\beta B + \beta J x_*), \quad (3.6)$$

which satisfies the same equation as m as we have seen. Therefore, we have $m = x_*$, which means that the long-range Ising model realises the mean field theory exactly.

4. Sheet 1, Problem 4*: First-Order and Tricritical Transitions

Study

$$f(m) = \alpha_2(T)m^2 + \alpha_4 m^4 + \alpha_6 m^6, \quad \alpha_4 < 0, \quad \alpha_6 > 0, \quad (4.1)$$

as $\alpha_2(T)$ decreases through zero. Sketch the changing free-energy landscape, prove that the first-order transition occurs at $\alpha_2 = \alpha_4^2 / (4\alpha_6)$, and find the magnetisation jump. Draw the phase diagram in the (α_2, α_4) plane. At $\alpha_4 = 0$ the transition at $\alpha_2(T_c) = 0$ is tricritical. Calculate its mean field exponents $\alpha, \beta, \gamma, \delta$; include a term proportional to Bm when finding the last two.

Solution. $f(m)$ crosses the m -axis at

$$m = 0, \quad m^2 = \frac{-\alpha_4 \pm \sqrt{\alpha_4^2 - 4\alpha_2\alpha_6}}{2\alpha_6}, \quad (4.2)$$

where the latter exist when $\alpha_2 < \alpha_4^2/(4\alpha_6)$. While the turning points of $f(m)$ is at

$$m = 0, \quad m^2 = \frac{-\alpha_4 \pm \sqrt{\alpha_4^2 - 3\alpha_6\alpha_2}}{3\alpha_6}, \quad (4.3)$$

where the latter exists when $\alpha_2 < \alpha_4^2/(3\alpha_6)$ before $f(m)$ crosses the axis. The two minima are degenerate at $\alpha_2 = \alpha_4^2/(4\alpha_6)$; below this value the nonzero minima are favoured. At coexistence, the magnetisation jumps from $m = 0$ to $m = \pm\sqrt{-\alpha_4/(2\alpha_6)}$.

At the tricritical point, we have

$$f(m) \approx -Bm + a(T - T_c)m^2 + \alpha_6 m^6. \quad (4.4)$$

The minima are at

$$m^4 = -\frac{2a}{6\alpha_6}(T - T_c), \quad (4.5)$$

and hence $\beta = 1/4$. The heat capacity is given by

$$c = \beta^2 \frac{\partial^2 \log Z}{\partial \beta^2}. \quad (4.6)$$

Plugging the temperature dependent minima into the partition function, we find

$$f(m_{\min}) \sim (T_c - T)^{3/2} \implies c \sim f'' \sim |T - T_c|^{-1/2}. \quad (4.7)$$

Hence, the critical exponent $\alpha = 1/2$. With the presence of an external source, the minima satisfies

$$B = 2a(T - T_c)m + 6\alpha_6 m^5 \implies m \sim B^{1/5} \quad (4.8)$$

which gives $\delta = 5$. Then,

$$\frac{1}{\chi} = \left. \frac{\partial B}{\partial m} \right|_{B=0} = 2a(T - T_c) + 30\alpha_6 m^4. \quad (4.9)$$

For $T > T_c$, $m = 0$, and hence $\gamma = 1$. For $T < T_c$, $m^4 \propto T_c - T$, so the two terms in $1/\chi$ are both linear in $|T - T_c|$ and $\gamma = 1$ below T_c as well.

5. Sheet 1, Problem 5: The XY Order Parameter

For a complex superfluid order parameter with

$$f(\psi) = \alpha_2 |\psi|^2 + \alpha_4 |\psi|^4, \quad (5.1)$$

sketch f when $\alpha_2 < 0 < \alpha_4$. Identify the possible ground states, explain spontaneous symmetry breaking here, and compare the mean field exponents α and β at $\alpha_2 = 0$ with those of the Ising model.

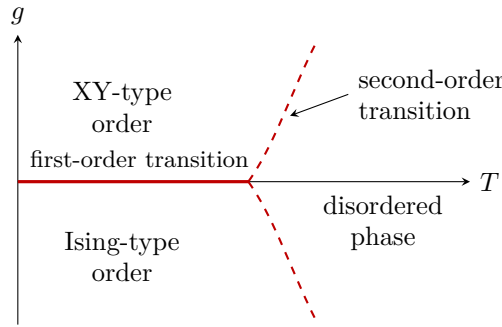
Solution. The free energy has a central maximum and a circle of degenerate minima. The ground states are $\psi = v e^{i\theta}$, where $v = \sqrt{-\alpha_2/(2\alpha_4)}$ and $0 \leq \theta < 2\pi$. Choosing one phase breaks the $U(1)$ symmetry of the free energy. As in the mean-field Ising model, $f_{\min} = -\alpha_2^2/(4\alpha_4)$ and $v \propto (T_c - T)^{1/2}$, giving $\alpha = 0$ and $\beta = 1/2$.

6. Sheet 1, Problem 6: Ising and XY Orders in an Anisotropic Spin Model

At each lattice site put a unit three-vector $\mathbf{s}_i = (s_i^x, s_i^y, s_i^z)$ and use

$$E = -J \sum_{\langle ij \rangle} \mathbf{s}_i \cdot \mathbf{s}_j + g \sum_i \left[(s_i^z)^2 - \frac{(s_i^x)^2 + (s_i^y)^2}{2} \right], \quad J > 0. \quad (6.1)$$

Use mean field reasoning to account for the phase diagram below,



Hint: Compare the ground states as g crosses zero at low T , then raise T separately on either side of $g = 0$. Detailed path-integral calculations are unnecessary.

Solution. The energy can be rewritten as

$$E = -J \sum_{\langle ij \rangle} \mathbf{s}_i \cdot \mathbf{s}_j + g \sum_i \left[\frac{3}{2} (s_i^z)^2 - \frac{1}{2} \right]. \quad (6.2)$$

When T is small and $g > 0$, to minimise the total energy, the system tends to align while the second term forces them into the $x - y$ plane. Therefore, it is a XY-type order. For $g < 0$, energy is minimised when all of the spins are in the z -direction and aligned. Hence, it is a Ising-type ordering. Therefore, going from $g < 0$ to $g > 0$ is a first-order phase transition.

As the temperature goes up, the system becomes disordered for both $g < 0$ and $g > 0$. Therefore, a second-order phase transition must have occurred.

7. Sheet 1, Problem 7: A Spatially Modulated Condensate

For a complex scalar field, consider

$$F = \int dx \left[\alpha_2(T) |\psi|^2 + \alpha_4 |\psi|^4 - \gamma \left| \frac{d\psi}{dx} \right|^2 + \kappa \left| \frac{d^2\psi}{dx^2} \right|^2 \right], \quad \alpha_4, \gamma, \kappa > 0. \quad (7.1)$$

Using a single nonzero Fourier mode $\psi_k = A_k e^{ikx}$ with $k = \pm k_0$, determine k_0 and show that the modulated phase appears at $\alpha_2 = \gamma^2/(4\kappa)$. Identify the symmetries broken in the ordered phase.

Solution. The free energy becomes

$$F = \int dx \left[\alpha_4 |A_k|^4 + (\alpha_2(T) - \gamma k^2 + \kappa k^4) |A_k|^2 \right]. \quad (7.2)$$

Then, we find $k_0 = \sqrt{\gamma/2\kappa}$. In this case, the free energy is

$$F = \int dx \left[\alpha_4 |A_k|^4 + \left(\alpha_2(T) - \frac{\gamma^2}{4\kappa} \right) |A_k|^2 \right], \quad (7.3)$$

and hence the phase transition happens at $\alpha_2 = \gamma^2/(4\kappa)$. A chosen nonzero mode breaks the global $U(1)$ symmetry and continuous spatial translations separately: a translation shifts its phase. A combined translation and compensating $U(1)$ rotation remains unbroken.

8. Sheet 1, Problem 8: Multicritical Scaling and Upper Critical Dimension

Take

$$f(m) = \alpha_2 m^2 + \alpha_{2n} m^{2n}, \quad n \in \mathbb{Z}_+. \quad (8.1)$$

Find β from $m \sim (T_c - T)^\beta$ in the ordered phase. Insert the result into the free energy to show that the heat-capacity exponent obeys $\alpha = 1 - 2\beta$. Compare this with the contribution to the free energy from fluctuations in the Gaussian path integral and establish the threshold $d_c = 2n/(n - 1)$ above which the mean field term dominates near criticality.

Solution. It can be easily obtained that $\beta = 1/(2n - 2) < 1$. Plugging back into the free energy and noting that $1 + 2\beta = 2n\beta$, we have

$$f \sim \alpha_2^{1+2\beta} \implies c \sim \alpha_2^{2\beta-1}, \quad (8.2)$$

and hence $\alpha = 1 - 2\beta$. In the Gaussian path integral, we have

$$c \sim |T - T_c|^{-2+d/2}. \quad (8.3)$$

Therefore, the critical dimension is

$$d_c = \frac{2n}{n - 1}. \quad (8.4)$$